

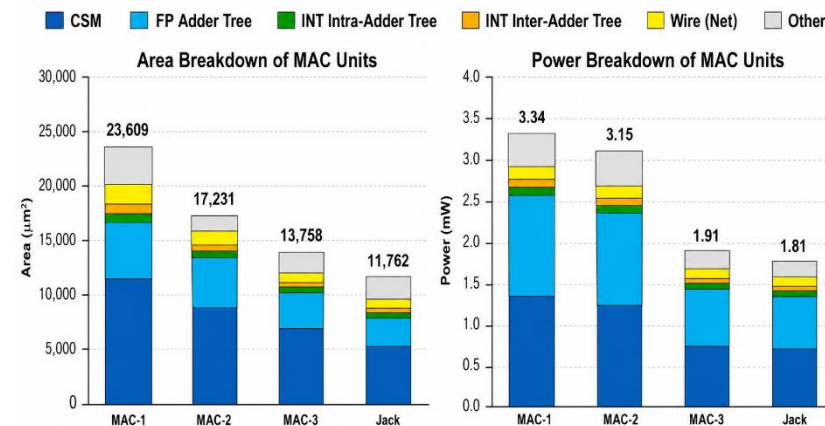
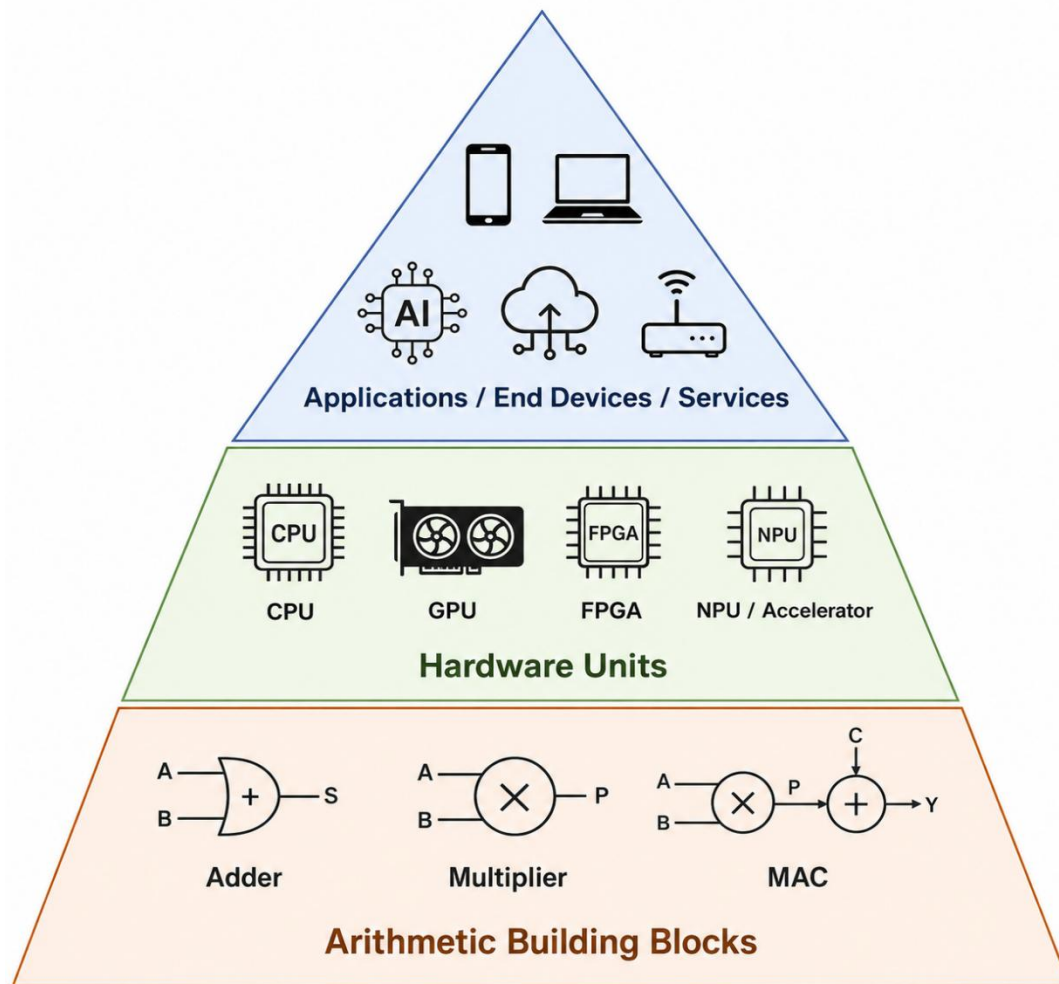
AXON: An Automated Netlist Optimization Framework for High-Speed Adders

Tiantian Yang, Xuanle Ren, Qingdian Wan, Qi Meng

ShanghaiTech University

yangtt2023@shanghaitech.edu.cn

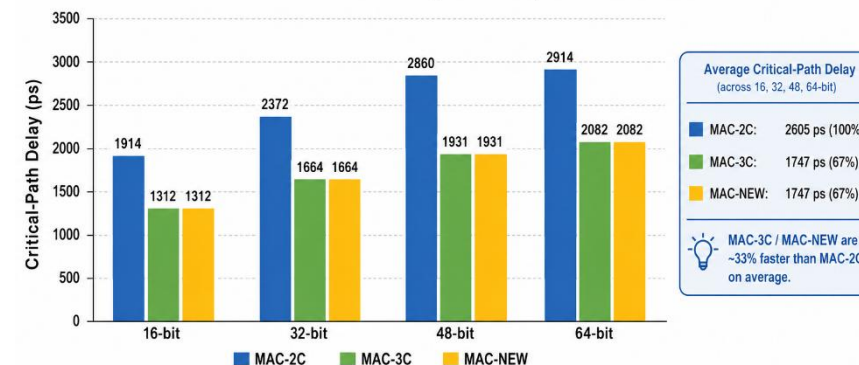
Background: Adders on Critical Path of Computing



Source: Redrawn from Noh et al., Jack Unit, 2025.

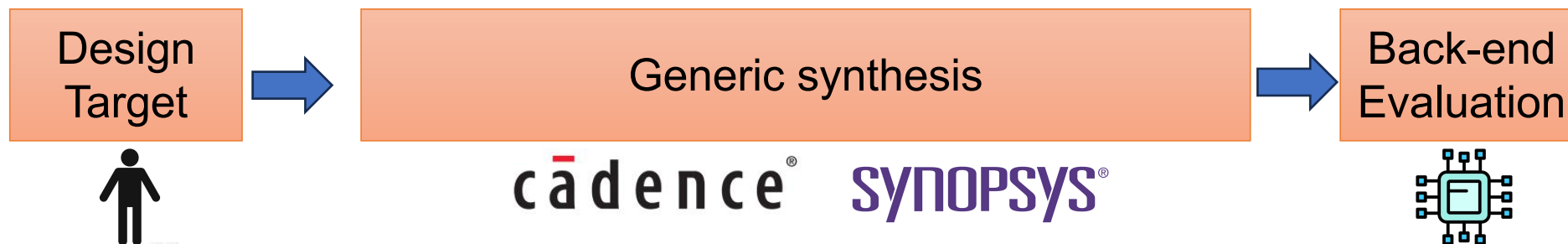
Performance Impact of MAC Architecture

Critical-Path Delay across Operand Sizes

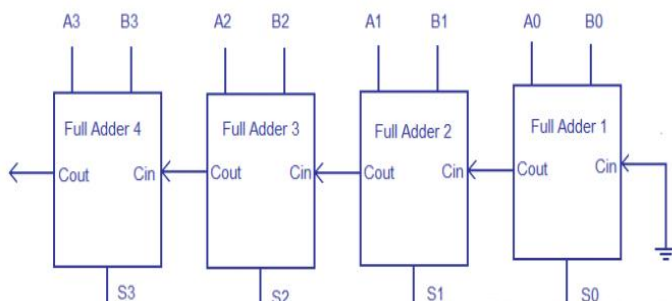


Source: Redrawn from Hoang et al., SoC 2009.

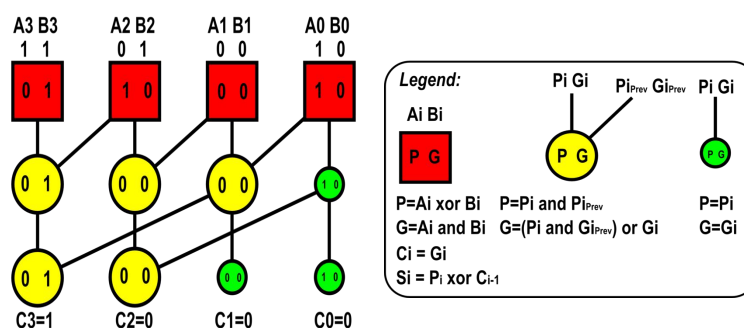
Problem: Generic Synthesis Is Not Adder-Specific



Ripple Carry



Parallel Prefix



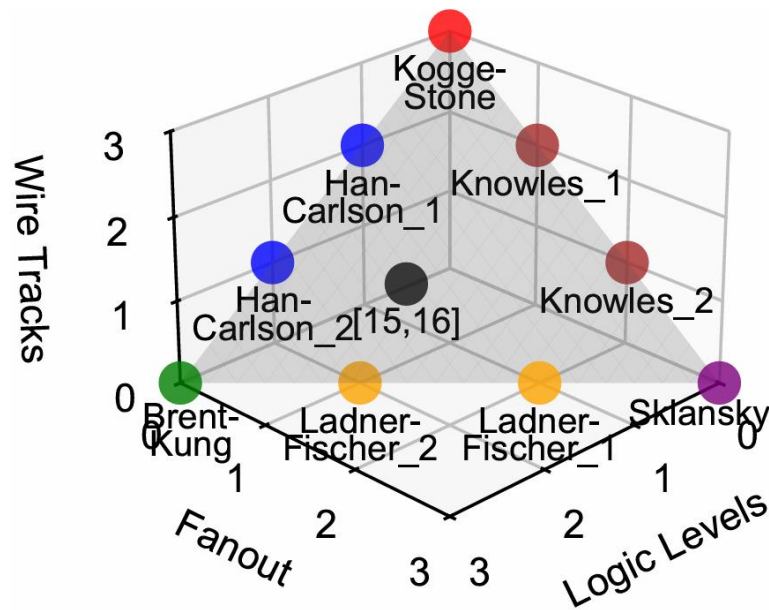
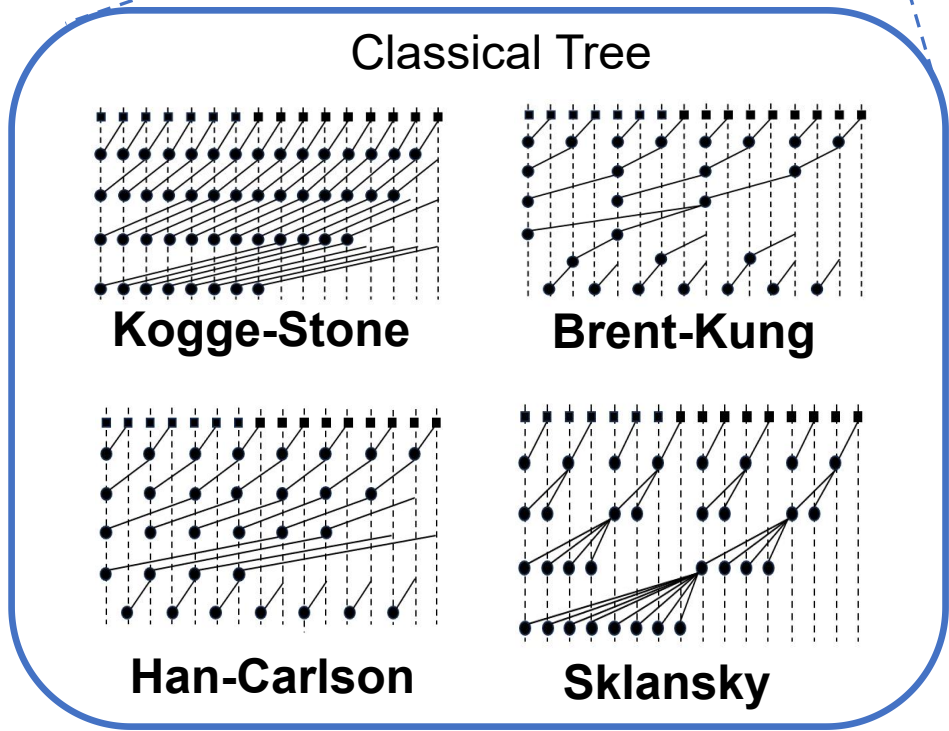
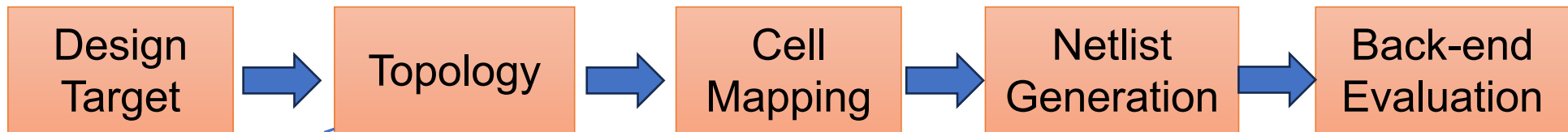
Gate Implementation

$$G = G1 + P1G0$$

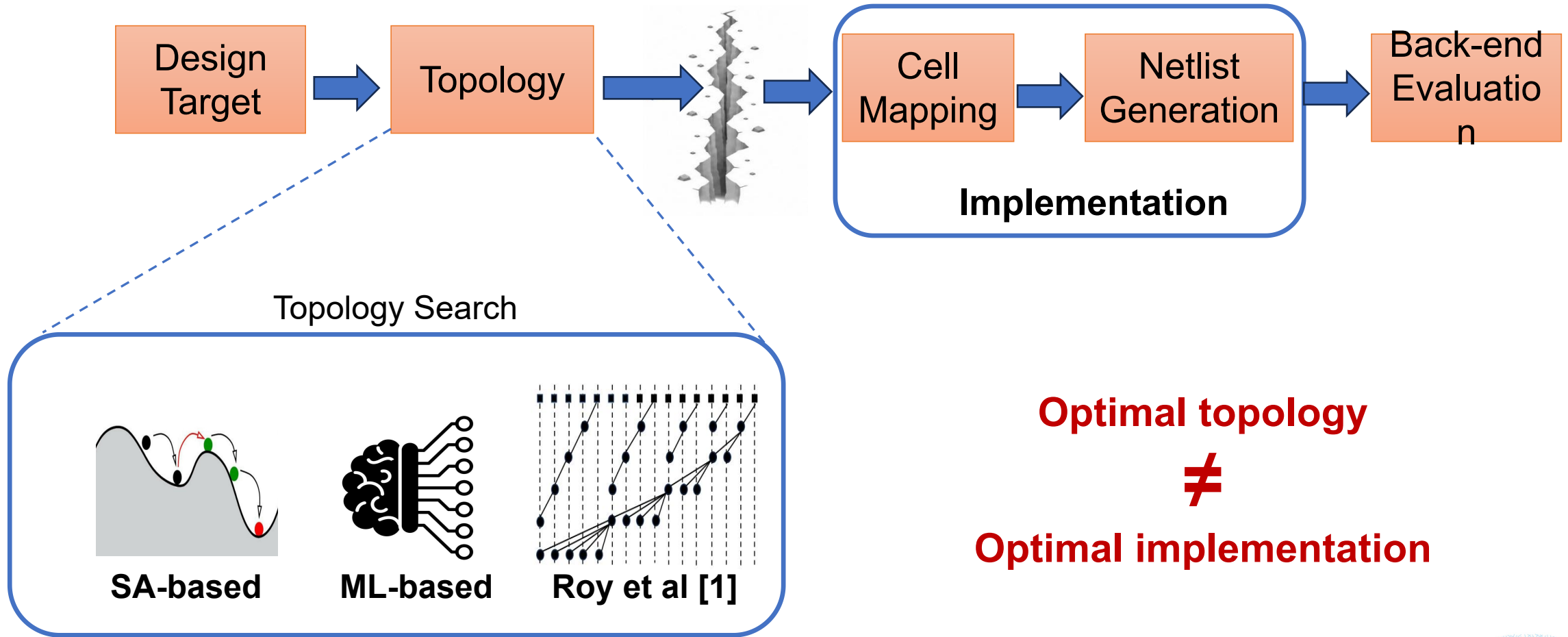
- AND + OR
- NAND/NOR
- AOI/OAI + INV



Problem: Limited Design Space In Classical Adders

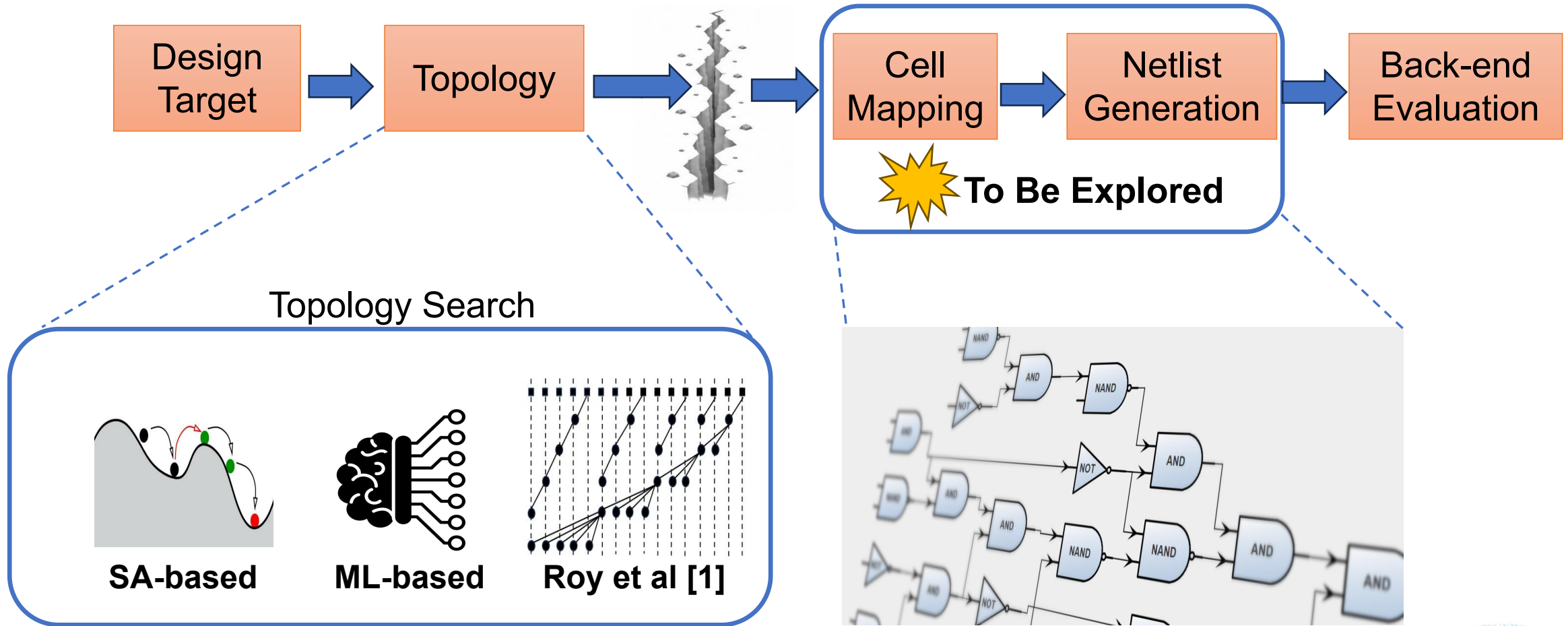


Problem: Existing Works Stop at Topology Search



[1]S. Roy, M. Choudhury, R. Puri, and D. Z. Pan, "Polynomial time algorithm for area and power efficient adder synthesis in high-performance designs," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 35, no. 5, pp. 820–831, 2016.

Problem: Less Explored Space on Netlist Level



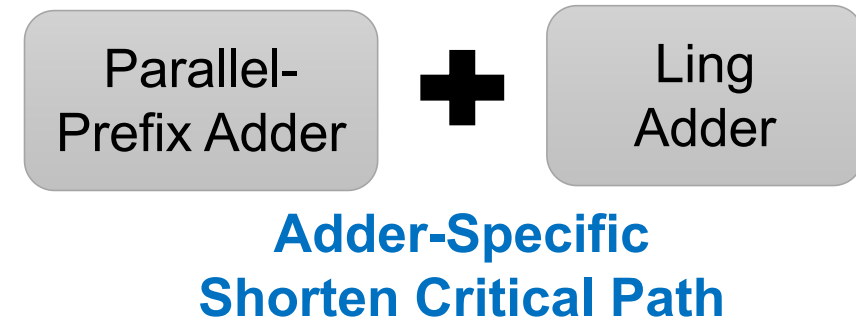
[1] S. Roy, M. Choudhury, R. Puri, and D. Z. Pan, "Polynomial time algorithm for area and power efficient adder synthesis in high-performance designs," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 35, no. 5, pp. 820–831, 2016.

Contributions of AXON

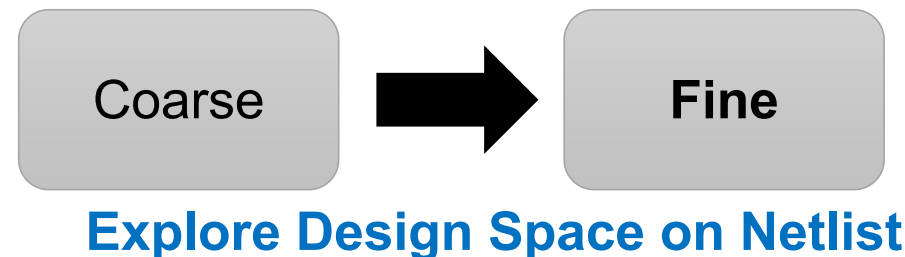
Topology-to-Netlist Optimization



Hybrid Prefix-Ling Topology

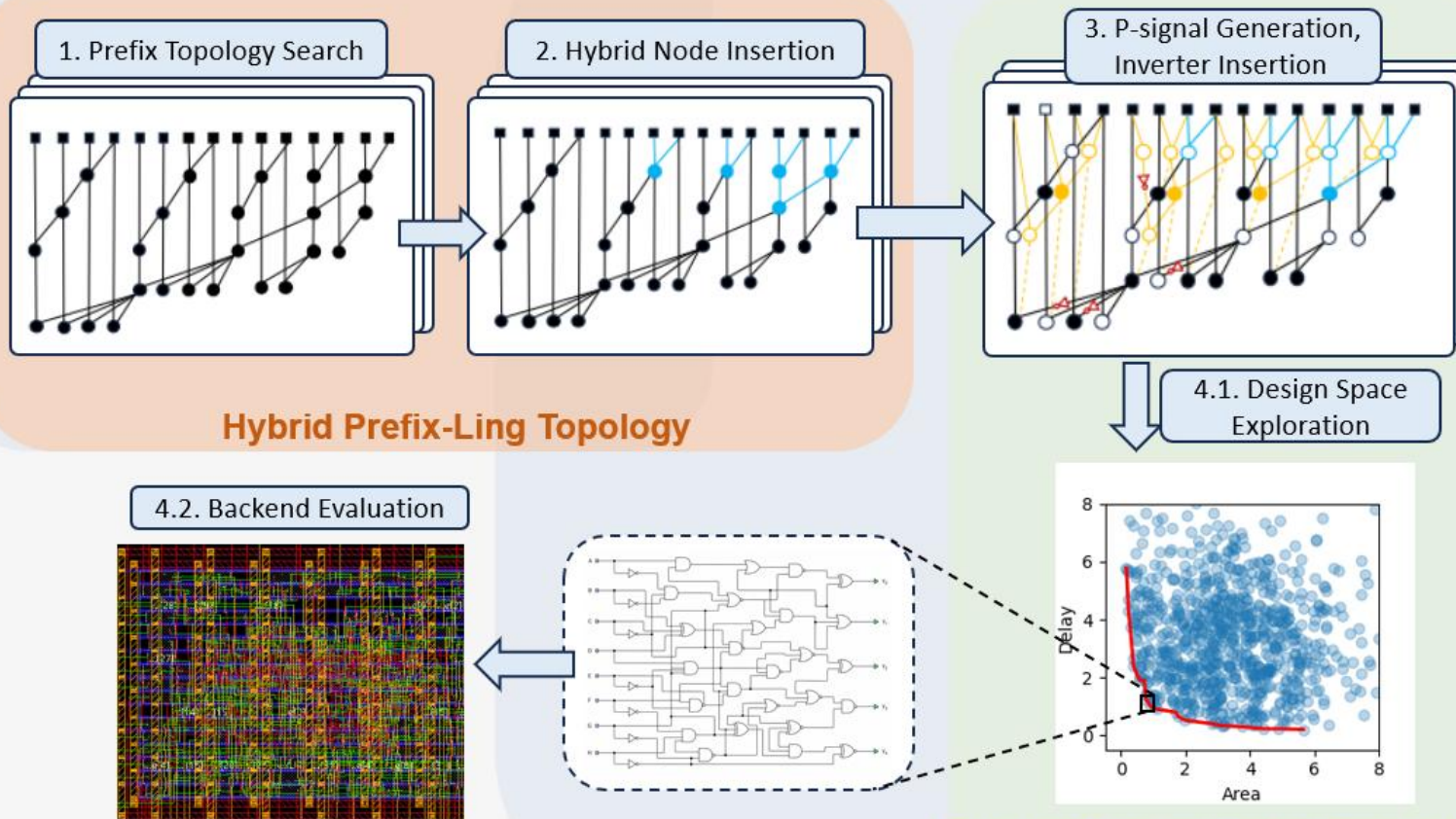


Efficient Design Space Exploration

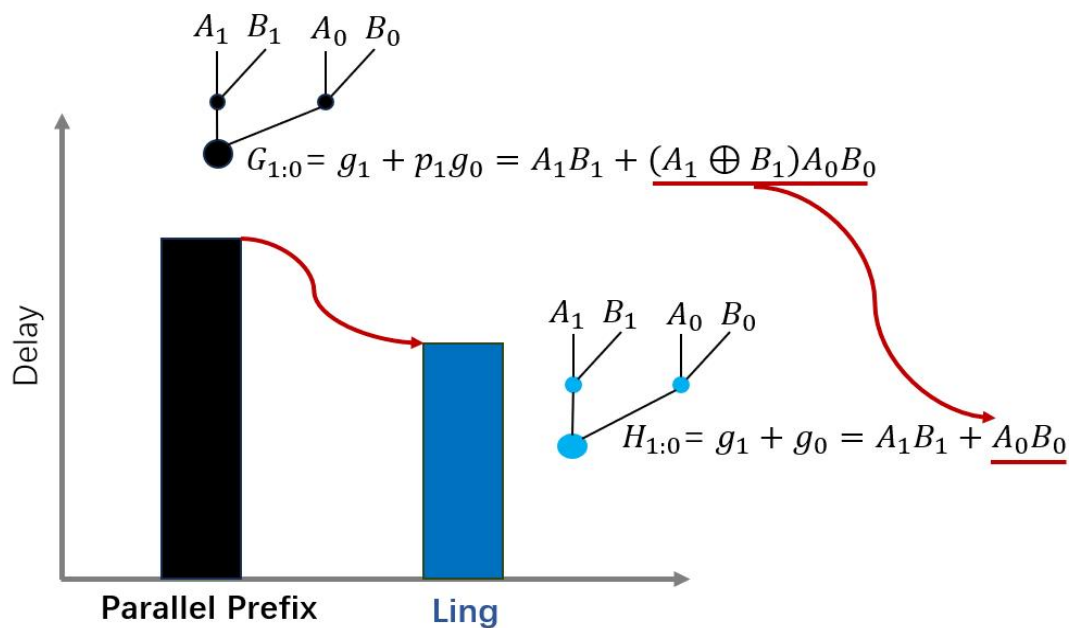


AXON Framework

Topology-to-Netlist Optimization



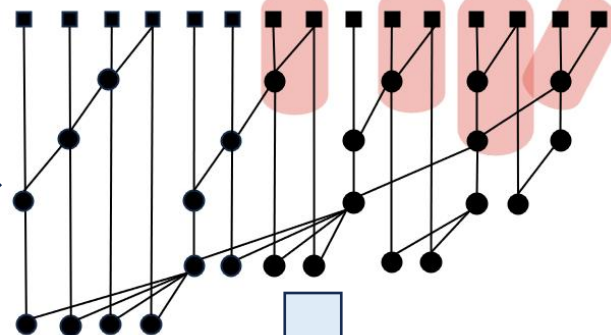
Hybrid Prefix-Ling Topology



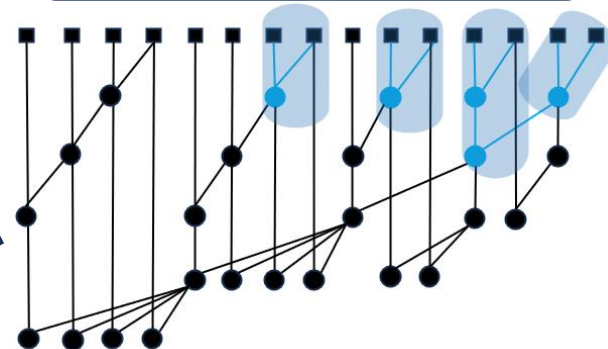
Multiple choices

Build a
parallel prefix
(minimal nodes)

Critical path prediction



Ling node transform



Benefits of Prefix-Ling:

Delay: -16.0%

ADP: -10.6%

Netlist Generation

P-Network Construction

Problem:

- P is complicated in Prefix-Ling
- P also affect PPA
- P was often tied to G

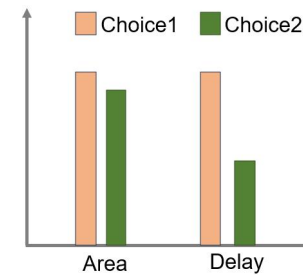
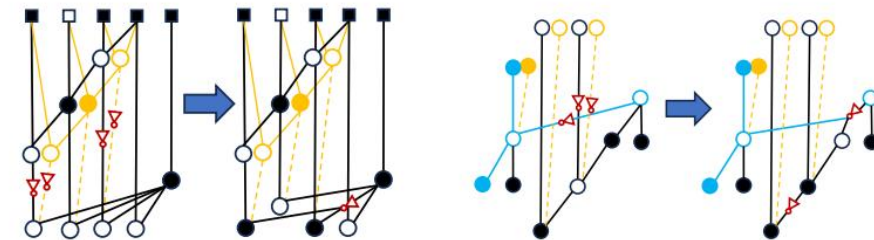
Solution:

Reuse the generated P nodes;
Build only indispensable ones.

Inverter Insertion

Problem:

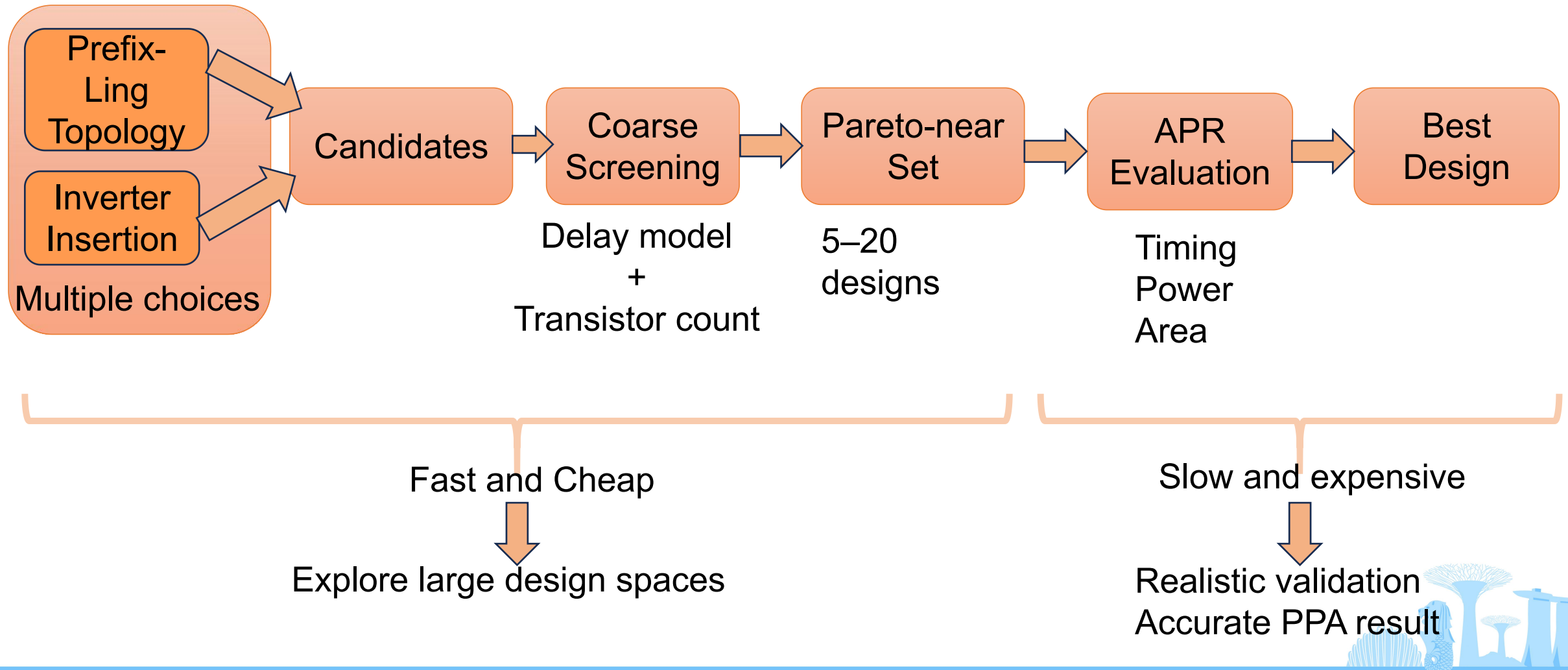
 Various insertion choices



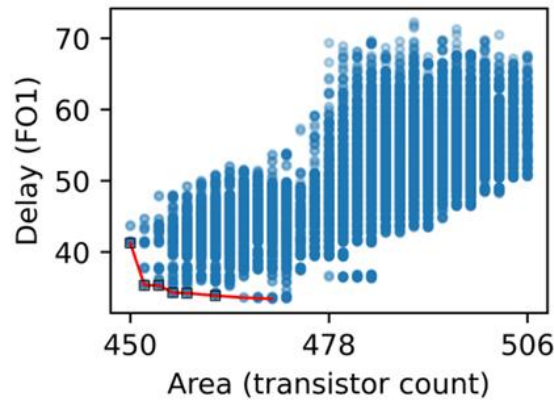
Same topology
Different INV insertion
↓
Different PPA



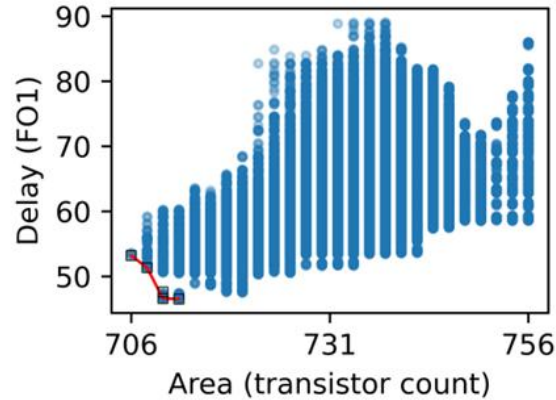
Hierarchical Design Space Exploration



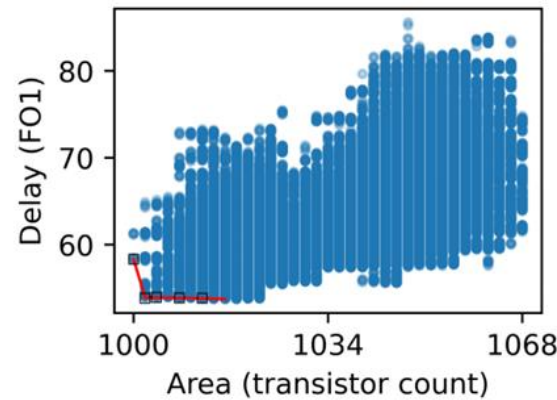
Results: AXON Advances the Pareto Frontier



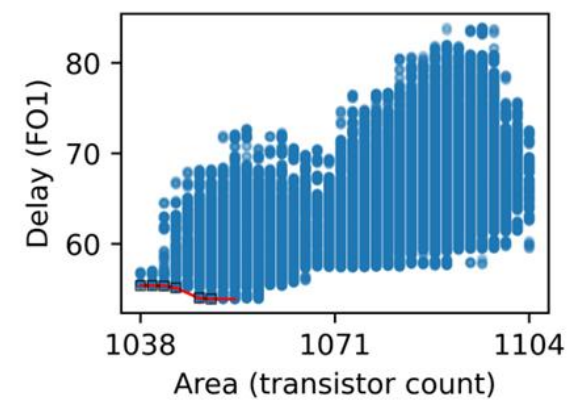
(a) 16-bit



(b) 23-bit



(c) 31-bit



(d) 32-bit

Key Findings

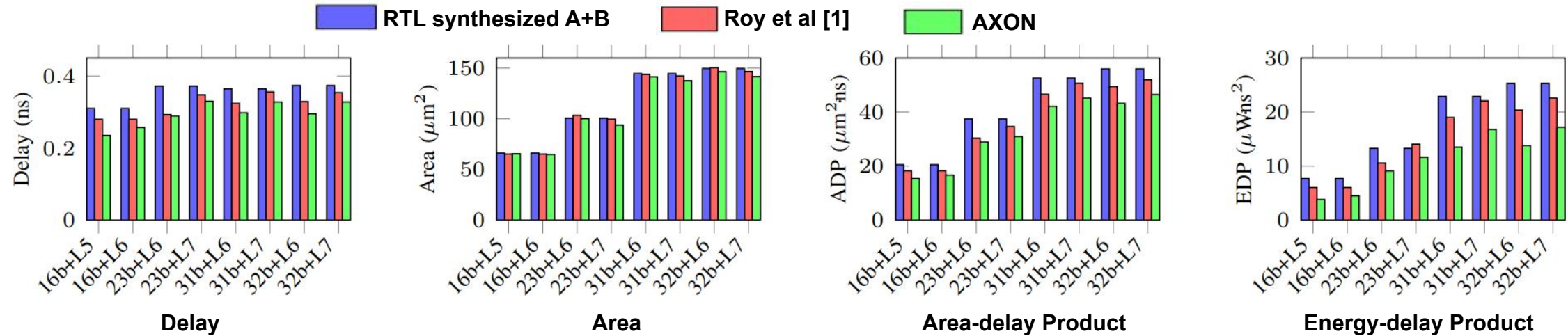
Same topology size, **different** netlist QoR.
area +6%, delay +56%(32 bit)

Implications:

Fixed topology $\neq$ fixed QoR. $\Rightarrow$ Netlist implementation strongly affect delay.
Space is large and nontrivial $\Rightarrow$ Coarse screening is necessary.
AXON exposes **hidden netlist-level design space**.
Justify **the need for architecture-to-netlist** co-optimization.



Results: Back-end Evaluation



vs Baseline1(RTL synthesized A+B)

Delay: up to -21%

ADP: up to -23%

EDP: up to -45%

vs Baseline2(Roy et al [1])

Delay: up to -10%

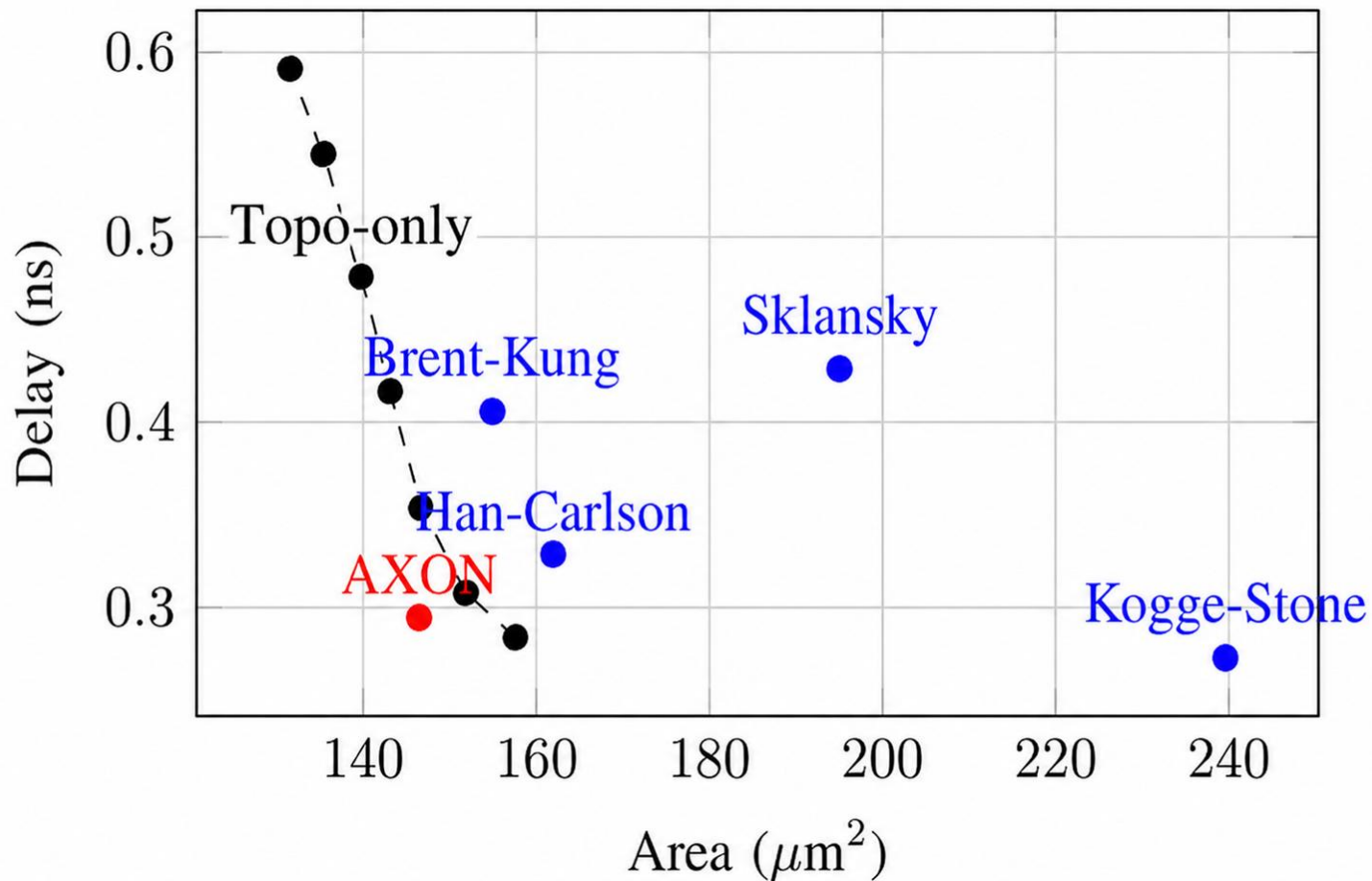
ADP: up to -13%

EDP: up to -32%

AXON outperforms both RTL-synthesized A+B and topology-only method.

[1]S. Roy, M. Choudhury, R. Puri, and D. Z. Pan, "Polynomial time algorithm for area and power efficient adder synthesis in high-performance designs," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 35, no. 5, pp. 820-831, 2016.

Results: AXON Push Pareto Frontier Downward



Conclusion & Takeaway

1. Adder optimization should not stop at topology search.
2. AXON bridges prefix structure and netlist implementation.
3. Structure-to-netlist optimization improves post-layout PPA.
4. Hierarchical evaluation makes large-space exploration practical.

Future Work:

Stronger physical awareness

Extension to broader arithmetic circuits(MAC units).



Thank you !



上海科技大学
ShanghaiTech University

